CHAPTER: HOW FORCES AFFECT MOTION

1. INTRODUCTION TO FORCE

In our daily lives, we observe objects resting, moving, speeding up, slowing down, or changing their direction. A stationary ball on a playground does not move on its own. To make it move, someone must push or kick it. Similarly, to stop a moving bicycle, the rider must apply brakes.

In physics, a **Force** is defined as a push or a pull acting upon an object resulting from its interaction with another object. Force is an external agency capable of altering a body's state of rest or uniform motion, or changing its shape and size.

Definition & Nature of Force
Force is a **vector quantity** because it possesses both magnitude and direction. It is represented by the symbol **F**.

SI Unit of Force = Newton (N) = kg·m/s²

CGS Unit: Dyne | **1 N = 10^5 Dynes**

Conceptualizing Push and Pull

- **Push:** Applying force away from oneself (e.g., pushing a door open, kicking a football).
- **Pull:** Applying force towards oneself (e.g., pulling a drawer, drawing water from a well).

2. EFFECTS OF FORCE

Force cannot be seen directly, but its presence and influence can be recognized entirely through its effects on objects. The primary effects of force include:

1. **Force can set a stationary body in motion:** Pushing a stationary cart causes it to move forward.
2. **Force can stop a moving body:** Applying brakes to a moving car brings it to a complete halt.
3. **Force can change the speed of a moving body:** Pressing the accelerator increases the vehicle's speed, whereas gentle braking slows it down.
4. **Force can change the direction of motion:** A batsman hits an oncoming cricket ball, altering its trajectory across the field.
5. **Force can change the shape and size of an object:** Squeezing a rubber ball, stretching a spring, or rolling dough alters physical dimensions.

EXAM POINTER!
Force does not always produce motion. For example, pushing a massive stone wall applies force, but because the wall does not move, no net acceleration or displacement occurs. Force can exist without causing actual displacement!

3. CONTACT AND NON-CONTACT FORCES

Forces are broadly categorized into two types based on whether physical contact is required between the interacting objects.

A. Contact Forces

Forces that act only when there is direct physical contact between the two interacting bodies are called **Contact Forces**.

- **Muscular Force:** The force exerted by the muscles of human or animal bodies (e.g., lifting a heavy bag, pulling a cart).
- **Frictional Force:** The opposing force that comes into play whenever two surfaces slide or attempt to slide across each other.
- **Normal Reaction Force:** The perpendicular contact force exerted by a surface on an object resting upon it.
- **Tension Force:** The pulling force transmitted through a string, rope, or cable when pulled tight from opposite ends.

B. Non-contact Forces

Forces that can act across a distance without any direct physical contact between interacting bodies are called **Non-contact Forces** or **Action-at-a-Distance Forces**.

- **Gravitational Force:** The attractive force existing between any two masses in the universe (e.g., Earth pulling objects toward its center).
- **Electrostatic Force:** The attractive or repulsive force between stationary electric charges (e.g., a charged comb attracting small pieces of paper).
- **Magnetic Force:** The force exerted by magnets on magnetic materials like iron or between magnetic poles.

Comparison Table: Contact vs. Non-contact Forces

Parameter	Contact Forces	Non-contact Forces
Physical Contact	Mandatory; surfaces must touch directly.	Not required; operates through fields across space.
Origin	Intermolecular interactions at touching surfaces.	Gravitational, electric, or magnetic fields.
Examples	Friction, Muscular Force, Tension, Normal Reaction.	Gravity, Electrostatic Force, Magnetic Force.
Distance Influence	Effective only during direct touch contact.	Magnitude decreases as distance increases (Inverse Square Law).

KEY LEARNING NOTE
Non-contact forces act through fields (e.g., Gravitational field, Electromagnetic field). Contact forces are essentially electromagnetic in origin at the atomic level due to electron repulsion between surface atoms!

4. BALANCED AND UNBALANCED FORCES

When multiple forces act simultaneously on a body, their combined overall effect depends on the net resultant force vector.

A. Balanced Forces

If the resultant of all forces acting on a body is zero ($\mathbf{F_{net} = 0}$), the forces are called **Balanced Forces**.

- Balanced forces **do not change** the state of rest or uniform linear motion of an object.
- They produce zero acceleration (a = 0).
- Balanced forces can change the **shape and size** of an object (e.g., squeezing a balloon equally from opposite sides).

B. Unbalanced Forces

If the resultant of all forces acting on a body is non-zero ($\mathbf{F_{net} \neq 0}$), the forces are called **Unbalanced Forces**.

- Unbalanced forces **always produce acceleration** or deceleration.
- They change either the speed, the direction of motion, or both.
- An object continues to accelerate as long as an unbalanced net force acts upon it.

Detailed Comparison Table

Property	Balanced Forces	Unbalanced Forces
Net Force (F_{net})	Equal to zero ($\sum F = 0$).	Greater than zero ($\sum F \neq 0$).
Effect on Motion	No change in speed or direction.	Causes change in speed, direction, or both.
Acceleration	Zero acceleration (a = 0).	Non-zero acceleration (a ≠ 0).
State of Object	Remains stationary or moves at constant velocity.	Accelerates or decelerates in direction of net force.
Example	Tug-of-war with equal teams pulling equally.	Tug-of-war where one team pulls harder than the other.

MUST-REMEMBER CONCEPT!
An object moving with a **constant velocity** has zero net force acting on it. Constant speed in a straight line implies balanced forces, NOT zero forces!

5. INERTIA AND NEWTON'S FIRST LAW

Galileo's Observation & Incline Experiment

Galileo Galilei deduced that objects move with constant speed when no force acts on them. Observing marble balls rolling down inclined planes, he noted that a ball accelerates coming down and decelerates going up. On a smooth horizontal plane, a ball should theoretically move infinitely with uniform velocity in the absence of external friction.

Definition of Inertia

Inertia is the inherent property of an object by virtue of which it resists any change in its state of rest or uniform motion in a straight line.

Mass is the quantitative measure of inertia: A heavier object has greater inertia than a lighter object. For example, pushing a massive stone requires significantly more effort than pushing a light plastic ball.

Types of Inertia

1. **Inertia of Rest:** The tendency of a body to remain at rest unless acted upon by an external force.
   Example: When a bus suddenly starts moving forward, passengers fall backward because their lower body moves with the bus while their upper body tends to stay at rest.
2. **Inertia of Motion:** The tendency of a body to continue in uniform motion unless stopped by an external force.
   Example: When a moving bus stops suddenly, passengers lurch forward because their body continues moving forward.
3. **Inertia of Direction:** The inability of a body to change its direction of motion without an external force.
   Example: When a car takes a sharp turn, passengers are thrown outward due to their body's desire to maintain its original straight trajectory.

Newton's First Law of Motion

Statement of Newton's First Law
"An object continues to be in a state of rest or of uniform motion in a straight line unless acted upon by an external unbalanced force."

$$\text{If } F_{net} = 0 \Rightarrow \text{Acceleration } (a) = 0$$

EXAM DEFINITION
Newton's First Law is also called the **Law of Inertia**. It provides a qualitative definition of force.

6. MOMENTUM AND NEWTON'S SECOND LAW

A. Momentum (p)

The impact produced by a moving object depends on both its mass and its velocity. A small pebble thrown gently produces little impact, but a small bullet fired from a gun possesses enormous stopping difficulty due to high speed. This combined quality is defined as **Momentum**.

Momentum Formula & SI Unit

Momentum (p) of a body is defined as the product of its mass (m) and velocity (v).

$$p = m \times v$$

Vector Quantity: Direction is the same as velocity direction.

SI Unit: kg·m/s (kilogram metre per second)

B. Newton's Second Law of Motion

The First Law states what happens when net force is zero. The Second Law quantitatively states what happens when an unbalanced external force acts on a body.

Statement & Mathematical Derivation

"The rate of change of momentum of an object is directly proportional to the applied unbalanced force and takes place in the direction in which the force acts."

Let mass = **m**, initial velocity = **u**, final velocity = **v** in time **t**.

Initial momentum (p_1) = m·u | Final momentum (p_2) = m·v

Change in momentum = Δp = m·v - m·u = m(v - u)

Rate of change of momentum = m(v - u) / t = m·a

$$F \propto m \cdot a \Rightarrow F = k \cdot m \cdot a \text{ (Where } k = 1)$$

$$F = m \times a$$

Definition of 1 Newton Force

One Newton (1 N) is defined as the amount of force required to produce an acceleration of 1 m/s² in a mass of 1 kg.

$$1 \text{ N} = 1 \text{ kg} \times 1 \text{ m/s}^2$$

PRACTICAL APPLICATION

A cricketer pulls his hands backward while catching a fast ball. By increasing the time (t) taken to stop the ball, the rate of change of momentum decreases, thereby reducing the impact force (F) on his hands!

7. NEWTON'S THIRD LAW OF MOTION

Forces never occur singly in nature; they always occur in equal and opposite pairs. Whenever one object exerts a force on another object, the second object instantly exerts an equal and opposite force on the first.

Statement of Newton's Third Law
"To every action, there is always an equal and opposite reaction."

$$F_{AB} = -F_{BA}$$

Where F_{AB} is the force exerted by body A on body B (Action), and F_{BA} is force exerted by body B on body A (Reaction).

Key Features of Action-Reaction Pairs

- Action and reaction forces act **simultaneously**.
- They act on **two different objects**, never on the same object. Therefore, action and reaction forces can *never cancel each other out*.
- Even though force magnitudes are equal, their accelerations may differ if masses are different (since a = F/m).

Real-Life Examples of Newton's Third Law

1. **Recoil of a Gun:** When a bullet is fired from a gun, the gun exerts a forward action force on the bullet. The bullet exerts an equal backward reaction force on the gun, causing it to recoil.
2. **Walking on the Floor:** We push the ground backward with our feet (Action), and the ground exerts an equal reaction force pushing us forward (Reaction).
3. **Rocket Propulsion:** Hot gases are expelled downward through the nozzle at high speed (Action), producing an upward reaction force that drives the rocket upward.
4. **Rowing a Boat:** Oars push the water backward (Action), and water pushes the boat forward (Reaction).

Law of Conservation of Momentum

According to Newton's Third Law, in an isolated system (where no external unbalanced forces act):

Conservation Equation
"The total momentum of a system of colliding objects remains constant/conserved before and after collision."

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

COMMON EXAM TRAP!
Why don't action and reaction cancel out to produce equilibrium? Because they act on **two different bodies**! Equilibrium requires opposite forces acting on the *same* body.

8. FRICTION AND GRAVITY

A. Friction (Frictional Force)

Friction is the contact force that opposes the relative motion between two surfaces in contact. It acts parallel to the contact surfaces in the direction opposite to motion.

Types of Friction:

- **Static Friction (f_s):** Self-adjusting force operating between resting surfaces preventing motion. Maximum value is called *Limiting Friction*.
- **Sliding Friction (f_k):** Opposing force when one object slides over another. (Sliding friction < Limiting friction).
- **Rolling Friction (f_r):** Opposing force when an object rolls over a surface. (Rolling friction is much smaller than sliding friction).

Friction: A Necessary Evil

Advantages of Friction	Disadvantages of Friction
Enables walking without slipping.	Causes wear and tear of machine parts and shoes.
Allows vehicles to stop when brakes are applied.	Leads to energy loss in the form of heat.
Enables writing with pen/chalk on paper/blackboard.	Reduces mechanical efficiency of machines.

B. Gravity and Gravitational Force

Gravity is the non-contact attractive force exerted by Earth pulling objects toward its center.

Weight vs. Mass
Mass (m): Quantity of matter contained in a body. SI Unit: **kg**. (Constant everywhere).
Weight (W): The force of gravity with which Earth attracts a body.

$$W = m \times g$$

Where **g = 9.8 m/s²** (Acceleration due to gravity on Earth). SI Unit of Weight: **Newton (N)**.

IMPORTANT DISTINCTION
Mass is a **scalar quantity** and remains constant anywhere in the universe. Weight is a **vector quantity** and varies according to local gravitational acceleration 'g'.

9. SOLVED NUMERICAL PROBLEMS

Step-by-step solved physics numericals according to Class 9 NCERT & CBSE exam pattern.

Numerical Problem 1

Question: Calculate the force required to impart a velocity of 30 m/s in 10 seconds to a car of mass 1500 kg starting from rest.

Solution:
Given: Mass (m) = 1500 kg, Initial velocity (u) = 0 m/s, Final velocity (v) = 30 m/s, Time (t) = 10 s.
First, calculate acceleration (a):
a = (v - u) / t = (30 - 0) / 10 = **3 m/s²**
Now apply Newton's Second Law (F = m × a):
F = 1500 kg × 3 m/s² = **4500 N**.
Answer: The required force is 4500 N.

Numerical Problem 2

Question: A force of 5 N produces an acceleration of 8 m/s² on mass m_1 and 24 m/s² on mass m_2. What acceleration would it produce if both masses are tied together?

Solution:
For m_1: $m_1 = F / a_1 = 5 / 8 = 0.625$ kg
For m_2: $m_2 = F / a_2 = 5 / 24 = 0.208$ kg
Total combined mass $M = m_1 + m_2 = (5/8) + (5/24) = (15 + 5)/24 = 20/24 =$ **5/6 kg**
Combined acceleration (a):
a = F / M = 5 / (5/6) = 5 × (6/5) = **6 m/s²**.
Answer: Combined acceleration is 6 m/s².

Numerical Problem 3

Question: What is the momentum of an object of mass 200 g moving with a velocity of 15 m/s?

Solution:
Given: Mass (m) = 200 g = 200 / 1000 = 0.2 kg | Velocity (v) = 15 m/s
Momentum (p) = m × v = 0.2 kg × 15 m/s = **3 kg·m/s**.
Answer: The momentum is 3 kg·m/s.

Numerical Problem 4

Question: A bullet of mass 20 g moving with a speed of 150 m/s strikes a sandbag and comes to rest in 0.05 seconds. Calculate the retarding force.

Solution:
Given: m = 0.02 kg, u = 150 m/s, v = 0 m/s, t = 0.05 s
a = (v - u) / t = (0 - 150) / 0.05 = **-3000 m/s²**
F = m × a = 0.02 kg × (-3000 m/s²) = **-60 N**.
Answer: The retarding force is 60 N (negative sign indicates retardation).

Numerical Problem 5 (Recoil Velocity of Gun)

Question: A bullet of mass 10 g is fired from a gun of mass 4 kg with a velocity of 400 m/s. Calculate the recoil velocity of the gun.

Solution:

Mass of bullet (m_1) = 10 g = 0.01 kg, Velocity of bullet (v_1) = 400 m/s

Mass of gun (m_2) = 4 kg, Recoil velocity of gun = v_2

Initial system momentum before firing = 0

By Law of Conservation of Momentum:

$m_1v_1 + m_2v_2 = 0 \Rightarrow (0.01 \times 400) + (4 \times v_2) = 0$

$4 + 4\ v_2 = 0 \Rightarrow 4\ v_2 = -4 \Rightarrow v_2 =$ **-1 m/s**.

Answer: The gun recoils with a velocity of 1 m/s backwards.

QUICK REVISION & MASTER FORMULA SHEET

Physical Quantity / Concept	Formula / Equation	SI Unit & Remarks
Force (F)	$F = m \times a$	Newton (N) or kg·m/s²
Momentum (p)	$p = m \times v$	kg·m/s (Vector quantity)
Change in Momentum (Δp)	$\Delta p = m(v - u)$	kg·m/s
Newton's Second Law	$F = (p_2 - p_1) / t = m(v - u)/t$	Rate of change of momentum
Newton's Third Law	$F_{AB} = - F_{BA}$	Action and reaction act on 2 bodies
Conservation of Momentum	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	Isolated system ($F_{ext} = 0$)
Weight (W)	$W = m \times g$	Newton (N) [g = 9.8 m/s²]

REVISION SUMMARY CHECKLIST

✔ Mass is constant; Weight changes with gravity.

✔ Action-Reaction forces never act on the same object.

✔ Unbalanced net force is mandatory for producing acceleration.

10. PRACTICE QUESTIONS FOR SELF-ASSESSMENT

A. Very Short Answer Questions (1 Mark)

1. Define the SI unit of force (1 Newton).
2. What is the momentum of a body of mass 'm' resting at a point?
3. Which has more inertia: a rubber ball or a stone of the same size? Explain why.
4. Name the force responsible for slowing down a bicycle when pedaling is stopped.

B. Short Answer Questions (2–3 Marks)

5. Differentiate between Contact and Non-contact forces with two examples each.
6. Why do passengers fall in the backward direction when a bus suddenly accelerates from rest?
7. Explain why a karate player can break a pile of tiles with a single blow of his hand.
8. State Newton's Third Law of Motion and illustrate it with the example of swimming.
9. Differentiate clearly between Mass and Weight of an object.

C. Long Answer & Numerical Questions (4–5 Marks)

10. State Newton's Second Law of Motion. Derive the formula **F = m × a** from it.
11. State and prove the Law of Conservation of Momentum using Newton's Laws.
12. A truck starts from rest and rolls down a hill with constant acceleration. It travels a distance of 400 m in 20 s. Find its acceleration and the force acting on it if its mass is 7 tonnes (1 tonne = 1000 kg).
13. A rifle of mass 3 kg fires a bullet of mass 30 g with a velocity of 300 m/s. Calculate the initial recoil velocity of the rifle.

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